H1 Math Prelim Solutions 2018

1)

Minimum turning point: $k-6>0$

$k>6$

$(k-6)x^2-5x=3x-k$

$(k-6)x^2-8x+k=0$

Discriminant > 0: $b^2-4ac>0$

$(-8)^2-4(k-6)k>0$

$64-4k^2+24k>0$

$k^2-6k-16<0$

$(k-8)(k+2)<0$

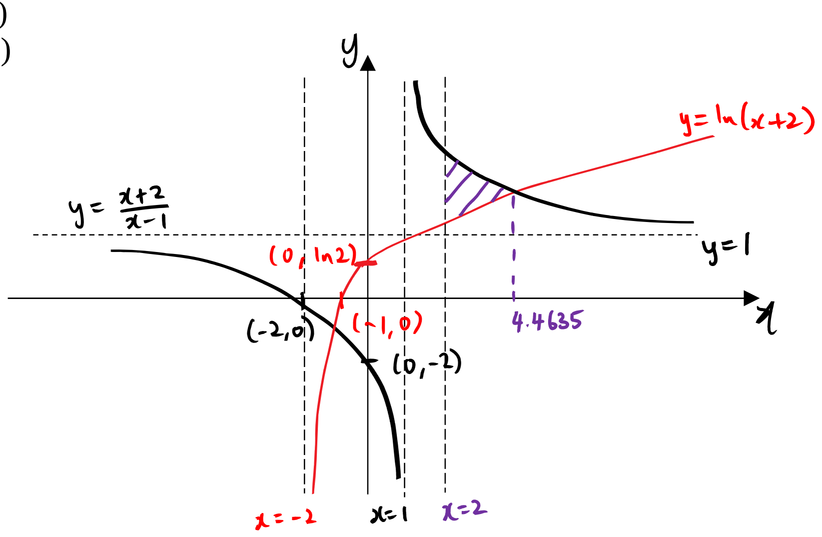
$-2<k<8$

Since $k>6$ and $-2<k<8 \Rightarrow 6<k<8$

$\{k\in \mathbb{R} : 6<k<8\}$

2)

(i)



(ii) From the GC, the point of intersection is (4.4635, 1.8662).

By GC, Area of the finite region $= \displaystyle\int_2^{4.4635}\left[\frac{x+2}{x-1}-\ln(x+2)\right]\mathrm{d}x = 2.14$

3)

(i) Height of ABC $=\sqrt{(5x)^2-(4x)^2}=3x$

$5xy\times 2+\frac{1}{2}(8x)(3x)\times 2=500$

$10xy+24x^2=500$

$y=\dfrac{500-24x^2}{10x}$

$$V=\frac{1}{2}(8x)(3x)y$$
$$=12x^2\left(\frac{500-24x^2}{10x}\right)$$
$$=\frac{6000x^2-288x^4}{10x}$$
$$=600x-\frac{144}{5}x^3$$

(ii)

$$\frac{\mathrm{d}V}{\mathrm{d}x}=600-86.4x^2$$

$$x^2=\frac{600}{86.4}=\frac{125}{18}$$

Since $x>0$, $x=2.6352$

	2.6352^-	0	2.6352^+
Sign of $\frac{\mathrm{d}V}{\mathrm{d}x}$	+	0	-
Slope of Tangent	/	-	\

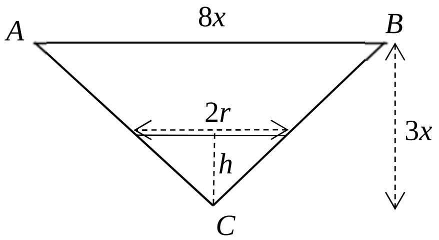
By sign test, V is maximum when $x=2.6352$.
When $x=2.6352$, $V=1054.09\approx1050$

(b) When $x=2$, $y=\frac{500-24(2)^2}{10(2)}=20.2$

` $V_w=\frac{1}{2}\times2r\times h\times20.2=20.2rh$

$3x=6$, $8x=16$

By similar triangles:



$$\frac{h}{6}=\frac{2r}{16}$$

$$r=\frac{8}{6}h$$

$$V_w=20.2\left(\frac{8}{6}h\right)h=\frac{404}{15}h^2$$

When $V_w=242.4$, $\frac{404}{15}h^2=242.4\Rightarrow h=3$

$$\frac{\mathrm{d}V_w}{\mathrm{d}h}=\frac{808}{15}h \quad , \quad \frac{\mathrm{d}V_w}{\mathrm{d}t}=10$$

$$\frac{\mathrm{d}h}{\mathrm{d}t}=\frac{\mathrm{d}h}{\mathrm{d}V_w}\times\frac{\mathrm{d}V_w}{\mathrm{d}t}$$

$$=\frac{15}{808(3)}\times10=0.0619$$

4)

(a)

$$\frac{5x+1}{(x+1)(2x+1)} = = \frac{A}{x+1} - \frac{B}{2x+1}$$

$$= \frac{A(2x+1) - B(x+1)}{(x+1)(2x-1)}$$

$$= \frac{(2A-B)x + A - B}{(x+1)(2x-1)}$$

$2A - B = 5$

$A - B = 1$

Solving simultaneously, $A = 4, B = 3$

$$\frac{5x+1}{(x+1)(2x+1)} = \frac{4}{x+1} - \frac{3}{2x+1}$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{5x+1}{(x+1)(2x+1)}\right) = \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{4}{x+1} - \frac{3}{2x+1}\right)$$

$$= -\frac{4}{(x+1)^2} + \frac{6}{(2x+1)^2}$$

(b)

$$\int \left(3x + \frac{2}{x^2}\right)^2 \mathrm{d}x = \int \left(9x^2 + \frac{12}{x} + \frac{4}{x^4}\right) \mathrm{d}x$$

$$= 3x^3 + 12 \ln x - \frac{4}{3x^3} + C$$

5)

(i) Let v, r and w represent the number of vanilla cupcakes, red velvet cupcakes and white chocolate cupcakes **<u>baked</u>** respectively.

$v + r + w = 400 - (1)$

$1.50v + 3.50r + 3.00w = 970 - (2)$

$\frac{5}{9}v(1.50) + \frac{3}{4}r(3.50) + \left(\frac{5}{9}v - 70\right)(3.00) = 450$

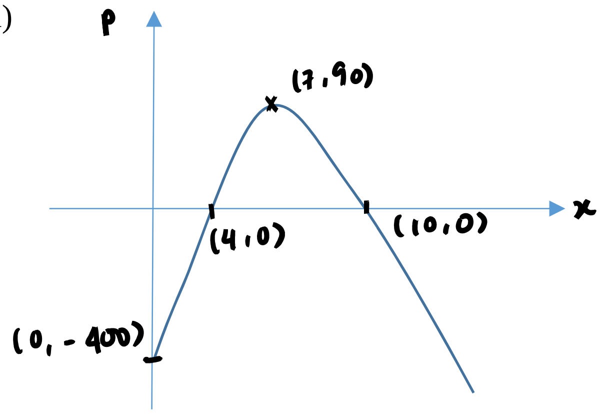
$\frac{5}{2}v + 2.625r = 660 - (3)$

By GC, $v = 180, r = 80, w = 140$

(ii) Profit = $450 - 180(0.60) - 80(2) - 140(1.80) = -70$

The store made a loss of $70 on Monday.

(iii)



(iv) The weight of cookies produced is 7 kg. The maximum value of P is \$90.

(v) It is the fixed cost of the stall.

6)

(i) $$\frac{\mathrm{P}(A \cap B)}{P(B)} = \frac{4}{7}$$

$$\mathrm{P}(A \cap B) = \frac{4}{7}P(B)$$

$$= \frac{4}{7}\left(\frac{7}{18}\right) = \frac{2}{9}$$

(ii) $$\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B) - \mathrm{P}(A \cap B)$$

$$1 - \mathrm{P}(A' \cap B') = \mathrm{P}(A) + \mathrm{P}(B) - \mathrm{P}(A \cap B)$$

$$1 - \frac{1}{3} = \mathrm{P}(A) + \frac{7}{18} - \frac{2}{9}$$

$$\frac{2}{3} - \frac{3}{18} = \mathrm{P}(A)$$

$$\mathrm{P}(A) = \frac{1}{2}$$

7)

(a) __ __ __ __ __ __

Number of usernames formed when repetitions are allowed $= 26^4 \times 10^2 = 45697600$

Number of codes with letter k exactly once and 2 different digits

$= (4\times1\times25\times25\times25)\times(10\times9) = 5625000$

Required probability= $= \dfrac{5625000}{45697600} \approx 0.123$

(b) Case 1: k __ __ __ __ __

Number of usernames formed with k as its first character but not 3 as its fifth character $= 1\times26\times26\times26\times9\times10 = 1581840$

Case 2: __ __ __ __ 3 __

Number of usernames formed with 3 as its fifth character but not k as its first character $= 25\times26\times26\times26\times1\times10 = 4394000$

Required probability $= \dfrac{1581840+4394000}{45697600} = \dfrac{17}{130} \approx 0.131$

8)

(i)

1) The probability that a pea seed germinates is the same for all 24 seeds.

2) A pea seed germinates independently of all other pea seeds.

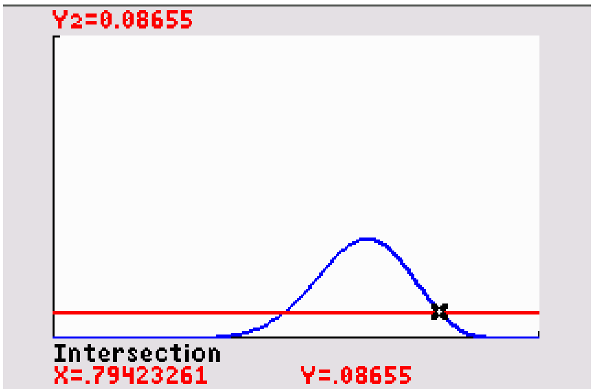
(ii)

Let X be the number of seeds that germinate out of a tray of 24.

$$X \sim \mathrm{B}(24, p)$$

$\mathrm{P}(X=15)+\mathrm{P}(X=16)=0.086550$

$\left({}^{24}C_{15}\right)(p)^{15}(1-p)^{9}+\left({}^{24}C_{16}\right)(p)^{16}(1-p)^{8}=0.086550$



By GC, $p=0.794233=0.7942$ or $p=0.476512$ (rej $\because p>0.5$)

(iii)

$\left[\mathrm{P}(X\geq20)\right]^{8}=\left[1-\mathrm{P}(X\leq19)\right]^{8}$

$=0.00121$

(iv) Let Y be the number of pea seeds that germinate out of 192 pea seeds in a carton.

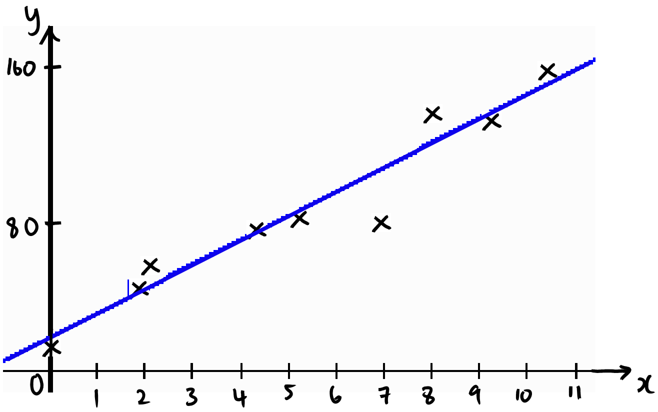
$$Y \sim \mathrm{B}(192, 0.794233)$$

$\mathrm{P}(Y\geq160)=1-\mathrm{P}(Y\leq159)=0.103$

(v) Answer in part (iv) is greater than the answer in (iii) because the event ‘each of the 8 trays in a carton contains at least 20 pea seeds that germinated’ is a subset of the event ‘a carton contains at least 160 pea seeds that germinated’.

9)

(i), (iii)



(ii) $r = 0.965$

(iii) $y = 12.579x + 17.940$

$y = 12.6x + 17.9$

(iv) The day with 6.9 hours of sunshine. The diagram shows that point (6.9, 78) is the furthest below the regression line as compared to the other points.

(v) $y = 12.579(9.5) + 17.940 = 137$

The particular day with 9.5 hours of sunshine where 190 ice cream cones were sold might not be in July. It might be a day in December which is the holiday season and more people visit the beach. Hence, it is not appropriate to use the regression line for estimation.

<u>OR</u>

The owner could have ran a sale on ice cream cones.

(vi) Since *x* and *y* are measured and we need to estimate *x* given *y*, regression line of *x* on *y* is used.

$x = 0.074099y - 0.96799$

$x = 0.074099(100) - 0.96799 = 6.44$

The estimate is reliable as $r = 0.965$ is close to 1, a linear model is appropriate and $y = 100$ is within the data range of $13 \le y \le 156$.

10)

(i) Let $y = x - 800$, so

$\sum y = -75.6, \qquad \sum y^2 = 1020.2$

$$\bar{x} = \bar{y} + 800 = \frac{-75.6}{50} + 800 = 798.488$$

$$s_x{}^2 = s_y{}^2 = \frac{1}{49}\left[1020.2 - \frac{(-75.6)^2}{50}\right] = 18.488 = 18.5$$

(ii) Let μ g be the population mean mass.

$H_0 : \mu = 800$

$H_1 : \mu \neq 800$

Level of significance: 5%

Test statistics: Since $n = 50$ is sufficiently large, by Central Limit Theorem, $\overline{X}$ is approximately normal.

When H_0 is true, $Z = \dfrac{\overline{X} - 800}{S / \sqrt{50}} \sim \mathrm{N}(0,1)$ approximately

Computation: $\overline{x} = 798.488$, $s^2 = 18.488$

$p - \text{value} = 0.0129$

Since $p - \text{value} = 0.0129 < 0.05$, H_0 is rejected at 5% level of significance. Hence, there is sufficient evidence that the supervisor's suspicion is valid.

(iii) It is not necessary to assume the mass of a packet of coffee powder follows a normal distribution. Since sample size = 50 is sufficiently large, by Central Limit Theorem, the sample mean mass of coffee powder ($\overline{X}$) is approximately normally distributed.

$\mathrm{H}_0 : \mu = 800$

$\mathrm{H}_1 : \mu > 800$

Level of significance: 5%

Test statistics: When H_0 is true, $Z = \dfrac{\overline{X} - 800}{18 / \sqrt{20}} \sim \mathrm{N}(0,1)$

Rejection region: $z \geq 1.6449$

Computation: $z = \dfrac{m - 800}{18 / \sqrt{20}}$

Owner's claim accepted

$\Rightarrow$ H_0 is rejected

$\dfrac{m - 800}{18 / \sqrt{20}} \geq 1.6449$

$m \geq 807$

$\therefore \{m \in \mathbb{R} : m \geq 807\}$

11)

Let X mins be the time taken by a boy and Y mins be the time taken by a girl.

$X \sim \mathrm{N}\left(11.51, 0.72^2\right)$

$Y \sim \mathrm{N}\left(13.17, 0.99^2\right)$

(i) $\mathrm{P}(X < 10) = 0.017987 = 0.0180$

(ii)

$\mathrm{P}(X < 9.5 \mid X < 10)$

$= \dfrac{\mathrm{P}(X < 9.5)}{\mathrm{P}(X < 10)}$

$= 0.146$

(iii)

$\mathrm{P}(X > t) < 0.4$

$1 - \mathrm{P}(X \le t) < 0.4$

$\mathrm{P}(X \le t) > 0.6$

$t > 11.7$

$\therefore \{t \in \mathbb{R} : t > 11.7\}$

(iv)

$\mathrm{E}\left(X_1 + X_2 - (Y_1 + Y_2)\right) = 2\mathrm{E}(X) - 2\mathrm{E}(Y) = -3.32$

$\mathrm{Var}\left(X_1 + X_2 - (Y_1 + Y_2)\right) = 2\mathrm{Var}(X) + 2\mathrm{Var}(Y) = 2.997$

$X_1 + X_2 - (Y_1 + Y_2) \sim \mathrm{N}(-3.32, 2.997)$

$\mathrm{P}(X_1 + X_2 \ge Y_1 + Y_2 + 1) + \mathrm{P}(X_1 + X_2 \le Y_1 + Y_2 - 1)$

$= \mathrm{P}\left(X_1 + X_2 - (Y_1 + Y_2) \ge 1\right) + \mathrm{P}\left(X_1 + X_2 - (Y_1 + Y_2) \le -1\right)$

$= 0.916$

(v)

$$\overline{W} = \frac{X_1 + X_2 + Y_1 + Y_2}{4}$$

$$\mathrm{E}\left(\overline{W}\right) = \frac{1}{4}\left[2\mathrm{E}(X) + 2\mathrm{E}(Y)\right] = 12.34$$

$$\mathrm{Var}\left(\overline{W}\right) = \frac{1}{16}\left[2\mathrm{Var}(X) + 2\mathrm{Var}(Y)\right] = 0.1873125$$

$\overline{W} \sim \mathrm{N}(12.34, 0.1873125)$

$\mathrm{P}\left(\overline{W} < 11.51\right) = 0.0276$